

Analysis In Competitions

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November 2024

1 CEZAR LUPO WORKSHEET

Problem 1. Let $0 < x_1 < 1$ and $x_{n+1} = x_n(1 - x_n)$ for $n \geq 1$. Show that

$$\lim_{n \rightarrow \infty} nx_n = 1.$$

Proof. By induction we find that $0 < x_n < 1$ for all $n \in \mathbb{N}$, which in turn proves that x_n is a bounded decreasing sequence. In particular it has a limit L , which satisfies the relation

$$L = L(1 - L)$$

or $L = 0$. Applying The Cezàro-Stolz theorem we then find:

$$\begin{aligned} \lim_{n \rightarrow \infty} nx_n &= \lim_{n \rightarrow \infty} \frac{n}{\frac{1}{x_n}} \\ &= \lim_{n \rightarrow \infty} \frac{(n+1) - n}{\frac{1}{x_{n+1}} - \frac{1}{x_n}} \end{aligned}$$

given that the second limit exists. Using the recursion formula, we find that

$$x_{n+1} - \frac{1}{x_n} = \frac{1}{x_n(1 - x_n)} - \frac{1}{x_n} = \frac{x_n}{x_n(1 - x_n)} = \frac{1}{1 - x_n}.$$

In particular we conclude that

$$\lim_{n \rightarrow \infty} \frac{(n+1) - n}{\frac{1}{x_{n+1}} - \frac{1}{x_n}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{1 - x_n}} = 1.$$

□

Problem 2. (i) let $(x_n)_n \geq 1$ be a sequence such that $x_1 \in (0, 1)$ and

$$x_{n+1} = x_n(1 - x_n^2).$$

evaluate the limit

$$\lim_{n \rightarrow \infty} \sqrt{n}x_n.$$

We claim that the limit is $\frac{1}{\sqrt{2}}$.

Proof. First note that an easy induction shows that $x_n \in (0, 1)$ for all $n \in \mathbb{N}$ and that the sequence is monotonically decreasing and thus convergent. Its limit fulfills the recursion that the x_n fulfill and thus equates to zero. Having this in mind, it suffices to evaluate

$$L := \lim_{n \rightarrow \infty} nx_n^2,$$

and then take the positive square root of the resulting limit. Using the Cesàro-Stolz theorem We may rewrite this limit as

$$L = \lim_{n \rightarrow \infty} \frac{n}{\frac{1}{x_n^2}} = \lim_{n \rightarrow \infty} \frac{n+1-n}{\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2}}.$$

Using the recursion for x_n we obtain the relation

$$\frac{1}{x_{n+1}^2} - \frac{1}{x_n^2} = \frac{1}{x_n^2(1-x_n^2)^2} - \frac{1}{x_n^2} = \frac{1-(1-x_n^2)^2}{x_n^2(1-x_n^2)^2} = \frac{2-x_n^2}{(1-x_n^2)^2} \xrightarrow{n \rightarrow \infty} 2.$$

Therefore. $L = \frac{1}{2}$ and the original limit is $\frac{1}{\sqrt{2}}$. $\square$

(ii) Let $(x_n)_{n \geq 1}$ be a sequence such that $x_1 = x > 1$ and

$$x_{n+1} = x_n + \sqrt{x_n} - 1, \quad n \geq 1.$$

Evaluate the limit

$$\lim_{n \rightarrow \infty} \frac{4x_n - n^2}{n \log n}$$

Problem 3. Consider the harmonic sequence $(H_n)_{n \geq 1}$, $H_n = \sum_{i=1}^n \frac{1}{i}$. Show that the sequence $x_n = \{H_n\}$ diverges. Here $\{x\}$ denotes the fractional part of x .

Proof. Clearly $\{H_n\} \in [0, 1)$, for all $n \in \mathbb{N}$, we will show a much stronger statement: the set of limit points of $\{H_n\}$ is all of $[0, 1]$. Suppose $q \in [0, 1]$, we construct a subsequence of $\{H_n\}$ which converges to q as follows. Since the Harmonic sequence is unbounded the difference of two terms is $\frac{1}{n}$, it enters every interval in of the form $[n, n+1]$ for $n \geq 1$. We claim that for each $k \in \mathbb{N}$, $(H_n)_{n \in \mathbb{N}}$ has a point in $H_{N_k} \in [k, n+q)$ such that $k+q-H_{N_k} \leq \frac{1}{k}$ or, in other words, $|q - \{H_{N_k}\}| \leq \frac{1}{k}$. Indeed, it is easy to see that if $j \in \mathbb{N}$ is the first natural number such that $H_j \in [n, n+1]$, then $j \geq n$. In particular the points of the sequence (H_n) which lie in $[n, n+1]$ all have distance $d \leq \frac{1}{n}$. Thus, if we choose N_k such that H_{N_k} is the last term of the sequence which comes before $n+q$, we find that $|n+q-H_{N_k}| \leq \frac{1}{n}$ or $|q - \{H_{N_k}\}| \leq \frac{1}{n}$. In this way we obtain a subsequence $(H_{N_k})_{k \in \mathbb{N}}$ with fractional part converging to q . $\square$

Remark. The same proof shows that we may replace H_n with any diverging sum $b_n = \sum_{k=1}^n a_k$ with converging arguments: $\lim_{n \rightarrow \infty} a_n = 0$.

Problem 4. Let $(a_n)_{n \in \mathbb{N}}$ be a decreasing sequence of positive reals such that that

$$\sum_{n=1}^{\infty} a_n < \infty.$$

Prove that

$$\lim_{n \rightarrow \infty} n a_n = 0.$$

Proof. Let $\epsilon > 0$ be arbitrary. Since $\sum_{n=1}^{\infty} a_n < \infty$ converges, there exist a $N \in \mathbb{N}$ such that $\sum_{i=N}^{\infty} a_n < \frac{\epsilon}{2}$. Then for all $j \geq 2N$, the monotonicity of $(a_n)_{n \in \mathbb{N}}$ allows us to obtain:

$$j a_j = \sum_{i=1}^j a_j \leq 2 \sum_{i=\lfloor \frac{j}{2} \rfloor}^j a_i \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this completes the proof. $\square$

Problem 5. (i) Evaluate the limit

$$\lim_{n \rightarrow \infty} \left(\frac{(1 + \frac{1}{n})^n}{e} \right)^n.$$

Before we determine the limit, we first mention a useful lemma.

Lemma 1.0.1. *Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be two convergent sequence with limits A and B respectively. Suppose the the sequence $(a_n)_{n \in \mathbb{N}}$ is strictly positive with positive limit. Then we have*

$$\lim_{n \rightarrow \infty} a_n^{b_n} = A^B.$$

Proof. Let $y_n := a_n^{b_n}$ and $x_n := \ln(y_n)$. Then we have

$$\lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} \exp(x_n) = \lim_{n \rightarrow \infty} \exp(b_n \ln(a_n)) = A^B.$$

$\square$

Now we are ready to determine the limit of the sequence. We claim that it is equal to $\frac{1}{\sqrt{e}}$.

Proof. First note that we can rewrite the sequence terms as follows

$$\left(\frac{(1 + \frac{1}{n})^n}{e} \right)^n = \left(1 + \frac{(1 + \frac{1}{n})^n - e}{e} \right)^{\frac{e}{(1 + \frac{1}{n})^n - e} \cdot \left(\frac{(1 + \frac{1}{n})^n - e}{e} \cdot n \right)}.$$

Using the fact that

$$\lim_{t \rightarrow 0} (1 + t)^{\frac{1}{t}} = e,$$

and the previous lemma, we can conclude:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{(1 + \frac{1}{n})^n - e}{e} \right)^{\frac{e}{(1 + \frac{1}{n})^n - e} \cdot \left(\frac{(1 + \frac{1}{n})^n - e}{e} \cdot n \right)} = \exp \left(\lim_{n \rightarrow \infty} \frac{(1 + \frac{1}{n})^n - e}{e} \cdot n \right).$$

We will now investigate the limit inside of the exponential, in fact we will only calculate

$$\lim_{n \rightarrow \infty} \left((1 + \frac{1}{n})^n - e \right) \cdot n.$$

Let us first investigate the inner term. Using Taylor expansions on $\ln(x+1)$ and $\exp(x)$ at 0, we find:

$$\begin{aligned} \left(1 + \frac{1}{n} \right)^n &= \exp \left(n \cdot \ln \left(1 + \frac{1}{n} \right) \right) \\ &= \exp \left(n \cdot \left(\frac{1}{n} - \frac{1}{2n^2} + R_{\ln,3} \left(\frac{1}{n^3} \right) \right) \right) \\ &= e \cdot \exp \left(-\frac{1}{2n} + n \cdot R_{\ln,3} \left(\frac{1}{n^3} \right) \right) \\ &= e \cdot \left(1 - \frac{1}{2n} + n \cdot R_{\ln,3} \left(\frac{1}{n^3} \right) + R_{\exp,3} \left(\frac{1}{2n} + n \cdot R_{\ln,3} \left(\frac{1}{n^3} \right) \right) \right) \end{aligned}$$

Therefore we have

$$\left(1 + \frac{1}{n} \right)^n - e = -\frac{e}{2n} + \tilde{R}(n), \text{ with } \lim_{n \rightarrow \infty} \frac{\tilde{R}}{\frac{1}{n^2}} = 0$$

In particular,

$$\lim_{n \rightarrow \infty} \left((1 + \frac{1}{n})^n - e \right) \cdot n = -\frac{e}{2},$$

and thus

$$\exp \left(\lim_{n \rightarrow \infty} \frac{(1 + \frac{1}{n})^n - e}{e} \cdot n \right) = \exp \left(-\frac{1}{2} \right) = \frac{1}{\sqrt{e}}.$$

□

(ii) Evaluate the limit

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \left(\frac{k}{k+1} \right) \right)^{e^{\frac{1999}{n}} - 1}.$$

Let $y_n := \left(\prod_{k=1}^n \left(\frac{k}{k+1} \right) \right)^{e^{\frac{1999}{n}} - 1}$ and $x_n := \ln(y_n)$. We determine the

limit of $(x_n)_{n \geq 1}$. We may rewrite the x_n as follows:

$$\begin{aligned} \ln \left(\left(\prod_{k=1}^n \left(\frac{k}{k+n} \right) \right)^{e^{\frac{1999}{n}} - 1} \right) &= \left(e^{\frac{1999}{n}} - 1 \right) \sum_{k=1}^n \ln \left(\frac{k}{k+n} \right) \\ &= \left(e^{\frac{1999}{n}} - 1 \right) \sum_{k=1}^n \ln \left(\frac{1}{1 + \frac{n}{k}} \right) \\ &= \left(e^{\frac{1999}{n}} - 1 \right) n \cdot \frac{1}{n} \sum_{k=1}^n \ln \left(\frac{1}{1 + \frac{n}{k}} \right). \end{aligned}$$

We can evaluate the limits of each factor individually, let us start with the left factor using Taylor expansion, we find

$$\left(e^{\frac{1999}{n}} - 1 \right) n = \left(1 + \frac{1999}{n} + R_2 \left(\frac{1999}{n} \right) - 1 \right) n,$$

which in turn implies

$$\lim_{n \rightarrow \infty} \left(e^{\frac{1999}{n}} - 1 \right) n = 1999.$$

Next we turn our attention to the right factor. If we define

$$f(x) := \ln \left(\frac{1}{1 + \frac{1}{x}} \right) = \ln \left(\frac{x}{x+1} \right),$$

then the left factor represents the n^{th} Riemann-sum of this function in the interval $[0, 1]$. In particular

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \left(\frac{1}{1 + \frac{n}{k}} \right) &= \int_0^1 \ln \left(\frac{x}{x+1} \right) dx. \\ &= \int_0^1 \ln(x) dx - \int_0^1 \ln(x+1) dx \\ &= [\ln x \cdot x - x]_0^1 + [\ln x \cdot x - x]_2^1 \\ &= -2 \ln 2. \end{aligned}$$

Finally we can conclude that

$$\lim_{n \rightarrow \infty} x_n = -3998 \cdot \ln 2,$$

which in turn implies

$$\lim_{n \rightarrow \infty} y_n = 2^{-3998}.$$

Remark. The key take away in these two problems was to use Taylor series expansions to estimate how fast certain things converge. Since comparing the growth of polynomials is easy, this made the rest quite doable. Furthermore the Riemann-sum trick used in problem (ii) is quite common; I was made painfully aware of this trick in the 2024 IMC, where I missed some easy, medal changing points.

Problem 6.

Problem 7. Suppose that $a_0 = 1$ and $a_{n+1} = a_n + \exp(-a_n)$ for $n \geq 1$. Does the sequence $a_n - \log n$ have finite limit.

Proof. First note that $(a_n)_{n \geq 0}$ is a monotonously increasing sequence and thus it takes on a well defined limit $L \in [1, \infty]$. If a_n were convergent, i.e. $L \in \mathbb{R}$, L would satisfy $\exp(-L) = 0$, which is not possible. In particular a_n is a divergent sequence to $+\infty$ and $\exp(-a_n)$ converges to zero.

Next we may rewrite the expression $a_n - \log n$ as $\log\left(\frac{\exp(a_n)}{n}\right)$. Since the function $\log x$ is continuous, it suffices to evaluate

$$\lim_{n \rightarrow \infty} \frac{\exp(a_n)}{n}.$$

Using the Cesàro-Stolz theorem, and the recursion formula, we find that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\exp(a_n)}{n} &= \lim_{n \rightarrow \infty} \frac{\exp(a_{n+1}) - \exp(a_n)}{1} \\ &= \lim_{n \rightarrow \infty} \frac{\exp(a_n + \exp(-a_n)) - \exp(a_n)}{1} \\ &= \lim_{n \rightarrow \infty} \exp(a_n)(\exp(\exp(-a_n)) - 1). \end{aligned}$$

It remains to evaluate the last limit. For this consider the function f given as

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto \frac{\exp(\exp(-x)) - 1}{\exp(-x)}.$$

Applying L'Hôpital's rule we see that

$$\begin{aligned} \lim_{x \rightarrow \infty} f(x) &= \lim_{x \rightarrow \infty} \frac{\exp(\exp(-x)) - 1}{\exp(-x)} \\ &= \lim_{x \rightarrow \infty} \frac{\exp(\exp(-x)) \cdot (-\exp(-x))}{-\exp(-x)} \\ &= 1. \end{aligned}$$

In particular, since a_n diverges to $+\infty$, we find that

$$\lim_{n \rightarrow \infty} \exp(a_n)(\exp(\exp(-a_n)) - 1) = 1.$$

This in turn guarantees that

$$\lim_{n \rightarrow \infty} (a_n - \log n) = 0.$$

□

Problem 8.

Problem 9. Does there exist a sequence $(a_n)_{n \geq 1}$ of positive reals such that

$$\sum_{i=1}^{\infty} \frac{1}{a_n}, \text{ converges and } , \prod_{i=1}^n a_n \leq n^n \text{ for all } n \geq 1?$$

There exists no such sequence.

Proof. Assume for sake of contradiction that $(a_n)_{n \geq 1}$ is such a sequence. Then, since all a_i are positive, the rearrangement theorem applies and

$$\sum_{i=1}^{\infty} a_{\sigma(n)}, \text{ converges,}$$

for all rearrangements of $\mathbb{N}$. Furthermore since $\sum_{i=1}^{\infty} \frac{1}{a_n}$ converges, we may rearrange the a_i to a sequence (a'_i) which is increasing. In particular we have $a'_i \leq a_i$ for all i . In particular, we find that

$$\prod_{i=1}^n a'_i \leq \prod_{i=1}^n a_i \leq n^n.$$

In particular $(a'_i)_{i \geq 1}$ is also a sequence of the desired type. Thus, we may assume without loss of generality, that the sequence a_i is increasing and we will do this from now on. Since $\frac{1}{a_i}$ is a decreasing sequence of positive reals such that $\sum_{i=1}^{\infty} \frac{1}{a_i} < \infty$, Oliver's lemma (Problem 4) applies:

$$\lim_{n \rightarrow \infty} \frac{n}{a_n} = 0.$$

Then, for $\epsilon := \frac{1}{e}$, there exists a $N_\epsilon \in \mathbb{N}$ such that $\frac{n}{a_n} < \frac{1}{e}$ for all $n \geq N_\epsilon$. Using this we find for sufficiently large n :

$$\begin{aligned} \prod_i^n a_i &= \prod_i^{N_\epsilon-1} a_i \cdot \prod_{i=N_\epsilon}^n a_i \\ &= C \cdot \prod_{i=N_\epsilon}^n a_i \quad (\text{where } C := \prod_i^{N_\epsilon-1} a_i) \\ &\geq C \frac{n!}{(N_\epsilon - 1)!} \cdot e^{n-N_\epsilon} \\ &= \hat{C} \cdot n! \cdot e^n \sim \hat{C} \cdot \left(\sqrt{2\pi n} \left(\frac{n}{e} \right)^n \right) \cdot e^n = C^* \cdot n^{n+\frac{1}{2}} \in \Theta(n^n \cdot \sqrt{n}). \end{aligned}$$

Where we used Stirling's formula for the asymptotic behavior. This immediately implies that the inequality $\prod_{i=1}^n a_i \leq n^n$ is not possible for all n . $\square$

Problem 10.

Problem 11. Let $(a_n)_{n \geq 1}$ be a decreasing sequence of positive and let $s_n = a_1 + \cdots + a_n$. Furthermore define

$$b_n := \frac{1}{a_{n+1}} - \frac{1}{a_n}.$$

Prove that if $(s_n)_{n \geq 1}$ is convergent then $(b_n)_{n \geq 1}$ is unbounded.

Proof. We will show the contrapositive: if $(b_n)_{n \geq 1}$ is bounded, then $(s_n)_{n \geq 1}$ is not convergent. Suppose that there exists a real number B such that

$$0 \leq \frac{1}{a_{n+1}} - \frac{1}{a_n} \leq B,$$

holds for all $n \in \mathbb{N}$. Without loss of generality we may choose B to be strictly positive since otherwise we may simply choose a larger B . Scaling the a_i by B we then get that $b_n \leq 1$ and this clearly does not change whether s_n is convergent or not. Using this we can then obtain

$$\frac{1}{a_n} - \frac{1}{a_1} = \sum_{i=2}^n \frac{1}{a_i} - \frac{1}{a_{i-1}} \leq n - 1.$$

In particular, there is a constant c such that

$$\frac{1}{a_n} \leq n + c.$$

However, this implies that s_n must be divergent since

$$s_n = \sum_{i=1}^n a_i \geq \sum_{i=1}^n \frac{1}{i + c} \rightarrow \infty.$$

□

Problem 12. Let $(x_n)_{n \geq 2}$ be a sequence of real numbers such that $x_2 > 0$ and

$$x_{n+1} = -1 + \sqrt[n]{1 + nx_n}.$$

(i) Prove that $\lim_{n \rightarrow \infty} x_n = 0$.

Proof. We first need the following lemma.

Lemma 1.0.2. Suppose $C \in [0, \infty)$, then

$$\lim_{n \rightarrow \infty} \sqrt[n]{n + c} = 1$$

Proof. by the AM-GM inequality we have

$$1 \leq \sqrt[n]{n + c} \leq \frac{2\sqrt{n + c} + (n - 2)}{n}$$

The Sandwich principle then implies the desired result.

□

We are now ready to prove the statement. A simple induction shows that x_n is a strictly positive sequence, we claim that it is also positive which would then imply that it is convergent. Indeed, recall that the Bernoulli-Inequality states

$$(1+x)^r \geq 1+rx$$

for $x \geq -1$ and $r \geq 0$, which is equivalent to

$$x \geq -1 + \sqrt[r]{1+rx}.$$

This, in turn, implies that

$$x_{n+1} = -1 + \sqrt[n]{1+nx_n} \leq x_n.$$

In other words, the sequence is bounded and decreasing and thus convergent. Suppose $L \geq 0$ is the limit. if L is not zero, then let $c := \frac{1}{L}$. We then obtain from the recursion formula that

$$\begin{aligned} L &= -1 + \lim_{n \rightarrow \infty} \sqrt[n]{1+nx_n} \\ &= -1 + \lim_{n \rightarrow \infty} \left(\sqrt[n]{x_n} \cdot \sqrt[n]{\frac{1}{x_n} + n} \right) \\ &\leq -1 + \lim_{n \rightarrow \infty} \sqrt[n]{c+n} \\ &= 0. \end{aligned}$$

Where we used the lemma for the last step. This implies that the limit was zero all along. $\square$

- (ii) Prove that $\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = 1$ and $\lim_{n \rightarrow \infty} nx_n = 0$.